
Working With Layers

Key Terms in this Chapter

Term	Description
Composite drawing	A drawing that contains multiple drawings created with the help of CADD layers.
Layer or level	A transparent surface created within a drawing.
Layer color	A color assigned to a layer.
Layer off	Making the drawing objects on a layer invisible.
Layer on	Making the drawing objects on a layer visible.
Layer line type	A line type assigned to a layer.
Locking a layer	Making the drawing objects on a layer permanent and non-editable.
Macro	A recorded set of instructions that can be used to repeat a task.

The Concept of Layers in CADD

Layers are imaginary transparent surfaces that can be created within a CADD drawing. You can draw on these imaginary surfaces and group drawing objects on different layers (Fig. 6.1). This helps to organize CADD drawings and makes editing much easier. You can turn the layers on or off to view or edit specific objects. When you turn a layer off, the objects on that layer disappear from the screen; when you turn it back on, they are displayed. Using layers enables you to control the display of specific information on the screen by turning a combination of layers on or off.

You may create dozens of layers in a drawing to segregate drawing objects. You may create separate layers to show dimensions, hatch patterns and for any drawing objects that are different in nature. This provides a great convenience in editing drawings. When drawings are complex and a number of drawing objects are overlapping, you can turn off the unwanted layers. This makes it much easier to select the desired objects for editing. When you are finished editing, you can turn the layers back on.

Layers are commonly used in professional architecture and engineering drawings. Architects and engineers often need to coordinate different engineering services in a plan. They can illustrate different engineering services on the same drawing using different layers. This helps them understand how one engineering service relates to another.

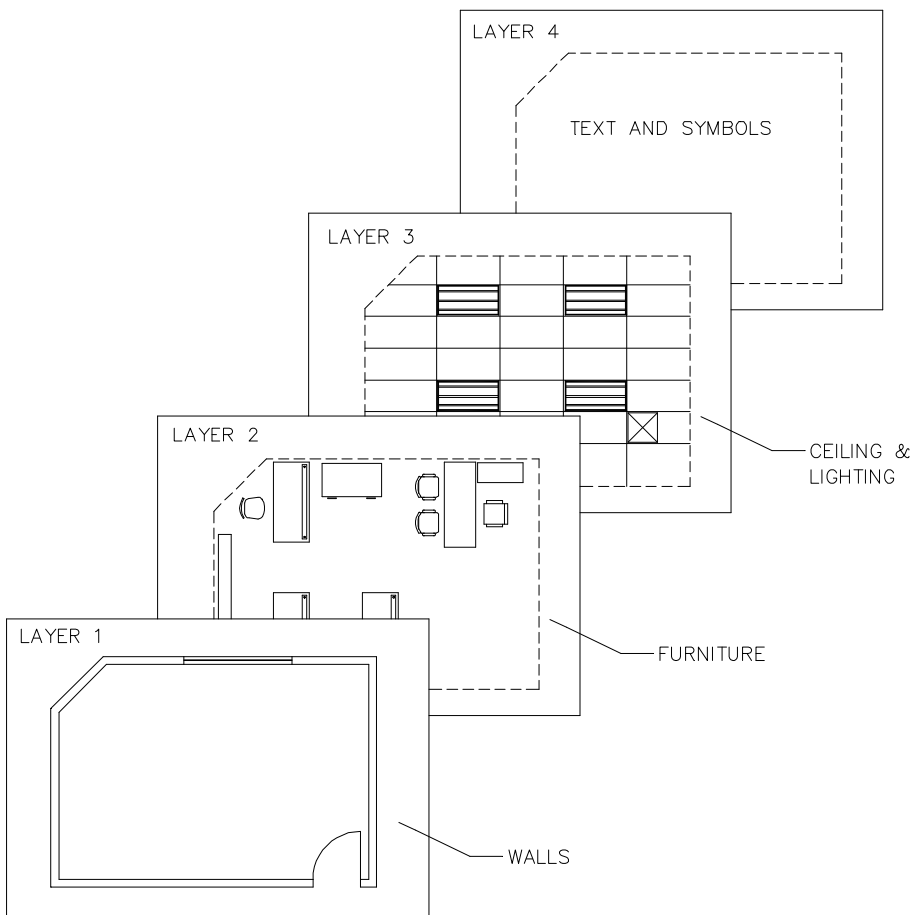


Fig. 6.1: *The concept of CADD layers.*

Creating Layers for an Engineer's Office

Fig. 6.1 illustrates how you can organize an engineer's office diagram in several layers. You can place walls and partitions on one layer, furniture and equipment on another, and so on for text, symbols, and construction lines, etc. You can also create a separate layer to show the ceiling and lighting plan, floor finish plan, or any other drawing objects that you want to include in the drawing.

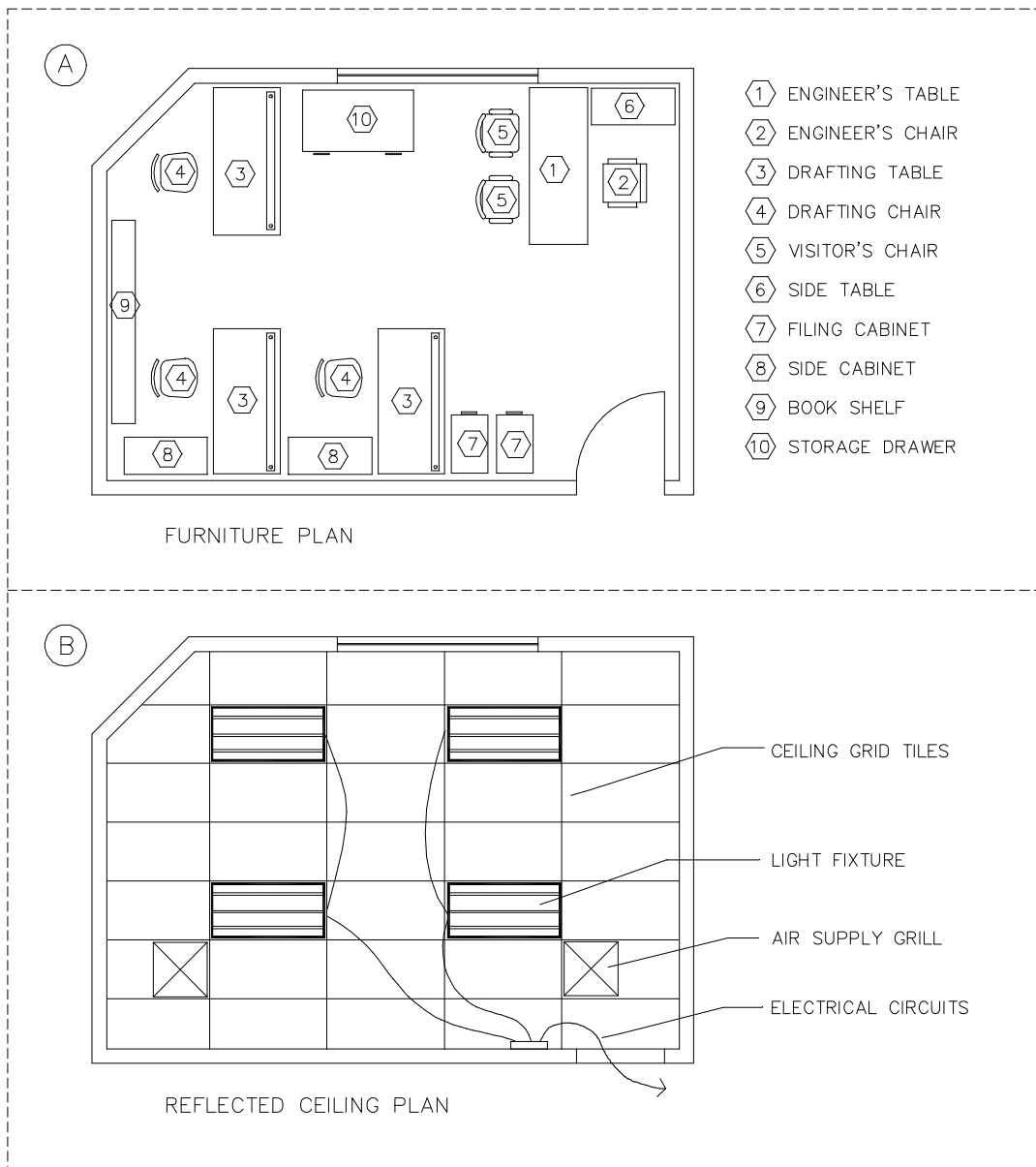


Fig. 6.2: Different drawings are created with the use of layers.

Fig. 6.2 shows how you can display a furniture plan and a reflected ceiling plan from this multi-layered drawing. When you need to display the furniture layout as shown in illustration A, you can turn on the wall and furniture layers, and turn off the rest. When you need to display the ceiling and lighting plan (Diagram B), you can turn on only the ceiling and the wall layers. You can turn the relevant text and symbol layers on and off as required.

Note:

Layers are given specific characteristics that distinguish them from each other. Each layer has a specific color and line type. Whatever you draw on a layer is drawn with that color and line type. This color coding helps to identify which drawing objects are placed on which layer.

The Tools to Work with Layers

CADD provides a number of tools that allow you to work efficiently with layers. Most programs use a layer control window that allows you to manage all aspects of working with layers.

The following are the basic tools for working with layers:

- Creating new layers
- Setting a layer to current
- Changing color and line type of layers
- Turning layers on or off
- Locking and unlocking layers

Creating New Layers

CADD allows you to create a number of layers in a drawing. To create a new layer you need to enter the name of the layer and assign a color and line type to it. The layer name should reflect the kind of information it contains.

Most companies follow standard guidelines in naming layers and placing specific information on specific layers. This allows CADD users to efficiently coordinate information. Professional institutes such as the American Institute of Architects (AIA) and the American Institute of Engineers (AIE) have developed standard naming conventions for layers that are commonly used by CADD professionals.

The following are some examples of layer names developed by the American Institute of Architects (AIA):

Layer name	Description
A-CLNG-GRID	Ceiling grid
A-WALL-EXTR	Exterior full height walls
E-LITE-CLNG	Ceiling-mounted light fixtures
E-LITE-CIRC	Lighting circuits
G-ANNO-NOTE	General notes and general remarks
G-ANNO-TTLB	Border and title block line work
I-FURN-TABL	Tables
I-FURN-SEAT	Chairs, sofas, etc.

AIA layers constitute hundreds of names that accommodate all disciplines of the building trade. The first letter of the layer name identifies a building trade. For example, A for architectural, E for electrical, I for interior design, S for structural, L for landscape and G for general drawings. There are dozens of layer names established for each engineering discipline.

For more information, visit the web site of the American Institute of Architects at <http://www.aia.org/>.

Facility Information Council (FIC) of the National Institute of Building Sciences (NIBS) has taken the initiative to develop universal CADD standards including layer names. For information visit their web site at: <http://www.nationalcadstandard.org/>.

Setting a Layer to Current

When you need to work on a specific layer, you must set that layer to be current by selecting it in the layer control window (Fig. 6.3). When a layer is set current, whatever you draw is drawn on that layer. The current layer name or number is displayed in the status bar.

Let's say you want to add some new furniture to the diagram as shown in Fig. 6.2. In order to maintain consistency, the new furniture must be drawn on the same layer as the existing furniture. You can set the furniture layer as the current layer. Subsequently, all the drawing work will be drawn on the furniture layer and should appear in the same color as the other furniture. If the color or line type is different, you may be working on the wrong layer.

Changing Color and Line Type of Layers

You can change the color and line type of layers as required. When you change the color and line type of layers, all the drawing objects on that layer are displayed with the new color and line type. You can make adjustments to colors and line types by selecting the layer from the layer control window (Fig. 6.3, 6.4) and choosing a color and line type for the layer.

Note: The colors used in a CADD drawing represent certain line weights and pen numbers when the drawings are plotted. More on colors and line weights is discussed in Chapter 8, "Printing and Plotting."

Turning Layers On or Off

You can turn any number of layers on or off by selecting them from the layer control window. To turn on a layer, highlight it in the layer control window and press the ON button (Fig. 6.3, 6.4). Similarly, you can turn the layers off when needed.

Turning layers on and off provides a great convenience for drawing and editing, as well as for plotting. To plot selected layers, simply turn off the rest of the layers. When drawing and editing, it is also a good idea to turn off unwanted layers, particularly when there is a lot of information on the screen. This speeds up the display process.

Locking and Unlocking Layers

Most CADD programs allow you to assign a lock or unlock status to layers. If a layer is locked, it means that you can't erase, move, or copy anything on that layer. To erase any drawing object from that layer, you must first unlock it. If there are certain drawing objects that are of a permanent nature and you don't want to run the risk of accidentally erasing or moving them, you can lock that layer.

Are Layers Absolutely Necessary?

Layers are not absolutely necessary, however they provide a convenient way to manage information in CADD drawings. For example, in the engineer's office example, without the help of this function you may have to create separate drawings to show the furniture plan, lighting plan, etc. With the help of layers, you can create just one composite drawing. The composite drawing can contain all the plans. It makes it easier to correlate information from one plan to another.

For complex drawings, layers are a must because they make editing much easier. Often you need to select only specific drawing objects for editing, and the use of layers makes it very easy to select them.